

End Semester Examination 2024

Name of the course: B.Tech

Name of the paper: Engineering Mathematics II

Time: 3 hours

Semester: II

Paper Code: TMA - 201

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b and c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each sub part carries 10 marks.

Q.1	(10×2 = 20 Marks)	
(a)	Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0..$	CO: 1
(b)	Solve by method of variation of parameters: $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}.$	
(c)	Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = x \cos x.$	
Q.2	(10×2 = 20 Marks)	
(a)	Express the following function in terms of unit step function and hence evaluate the Laplace transform $f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3 \end{cases}$	CO: 2
(b)	Find the inverse Laplace transform of $\frac{2s^2 + 5s - 4}{s^3 + s^2 - 2s}.$	
(c)	Solve the differential equation using Laplace transform method $\frac{d^2x}{dt^2} + 9x = \cos 2t, \quad x(0) = 1, \quad x\left(\frac{\pi}{2}\right) = -1.$	
Q.3	(10×2 = 20 Marks)	
(a)	Find the partial differential equation by eliminating arbitrary constants $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$	CO: 3
(b)	By using the Lagrange's method of multipliers find the general solution of $x(z^2 - y^2)\frac{\partial z}{\partial x} + y(x^2 - z^2)\frac{\partial z}{\partial y} = z(y^2 - x^2)$	
(c)	By using Charpit's method solve the partial differential equation: $px + qy = pq.$	

Q.4	(10×2 = 20 Marks)	
(a)	Solve the following equation by the method of separation of variables $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$ given that $u = 0$ when $t = 0$ and $\frac{\partial u}{\partial t} = 0$ when $x = 0$.	CO: 4
(b)	If a string of length l is initially at rest in equilibrium position and each of it's points is given the velocity $\left(\frac{\partial y}{\partial x}\right)_{t=0} = b \sin^3 \frac{\pi x}{l}$ find the displacement $y(x, t)$.	
(c)	Find the temperature in a bar of length 2 whose ends kept at zero and lateral surface insulated if initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$.	
Q.5	(10×2 = 20 Marks)	
(a)	Expand the function $f(x) = x \sin x$, as a Fourier series in the interval $-\pi \leq x \leq \pi$.	CO: 5
(b)	Find the Fourier sine and cosine transforms of $f(x) = e^{-ax}$.	
(c)	Obtain the Fourier series expansion of $f(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0 \end{cases}$	